

A Retroactive Contact Channel for Behavior Cloning on Low-Cost Arms

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Abstract: Low-cost servo arms ship without force sensing, and standard imitation-learning stacks observe joint positions only, so learned policies close the gripper open-loop against contact. For logs that record synchronized joint-position commands and measured positions, the servo tracking error $\delta = a[t-1] - s[t]$ is a contact proxy that can be computed retrospectively, and current stacks discard it. In a pre-registered six-design study at 280 demonstrations on a simulated force-critical task (5 seeds $\times$ 100 paired episodes), every observation that carries this channel explicitly outperforms the channel-free base, while predicting δ as an auxiliary training target does not. The strongest design appends a free-motion-compensated residual, the *lag excess*, and is the only one to outperform an action-history baseline at our pre-declared seed resolution (26.6% success versus 13.8%; 95% CI on the difference $[+2.8, +22.6]$). On 546 episodes across 16 public teleoperation datasets, the pre-registered prediction that successful grasps carry higher δ fails: under the frozen outcome proxy, high δ at grasp time is associated with failure, consistent with operators forcing jammed grasps. Code and data at [URL].

Keywords: Imitation Learning, Force Estimation, Low-Cost Manipulation

1 Introduction

Low-cost servo arms and open imitation-learning stacks have made manipulation research broadly accessible: ACT on ALOHA-class hardware [1], diffusion policies [2], and the LeRobot ecosystem [3] with its VLA descendants [4, 5, 6, 7] all train behavior cloning on teleoperated demonstrations, and all share one proprioceptive default: the current measured joint positions, one timestep, nothing else (Sec. 2).

This default fails at contact. The arms carry no force or tactile sensors, so a position-only policy cannot distinguish a seated grasp from an empty close; the failures are crushed objects, unseated grips, and drops during transport. The information needed to distinguish these cases is present in the logs. The servos are proportional position controllers, so when the jaw meets an object the measured position stalls while the command deepens, and the gap reflects the servo’s deflection under load. The authors of ACT state the physics directly: force “is implicitly defined by the difference between” leader and follower positions [1], and a recent intervention study shows that removing this discrepancy signal from ACT’s action labels breaks force-critical tasks [8]. Yet the stack keeps one operand of that subtraction in the action labels and never provides the difference as an observation; the LeRobot Feetech driver reads only `Present.Position` into the state vector, and the reference ACT implementation raises on `n_obs.steps != 1` [3].

The channel has a property no current- or torque-based estimator shares: it is recoverable retroactively. Wherever a recorded dataset’s action field holds joint-position commands synchronized with `observation.state` in a common frame, $\delta = a[t-1] - s[t]$ can be computed after the fact;

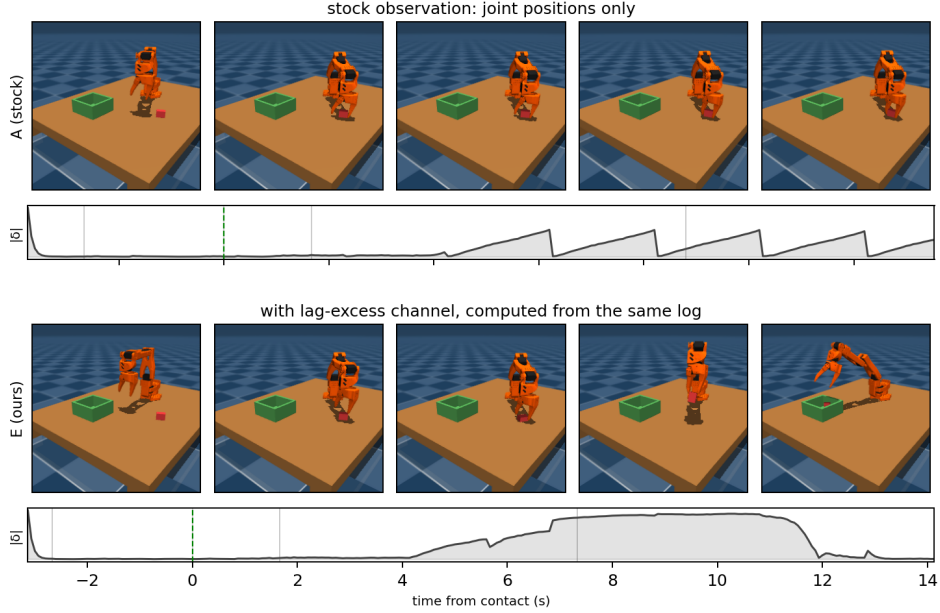


Figure 1: Two policies, identical scene and data budget (280 demonstrations). Top: the stock observation (joint positions only) reaches, grips, and releases without lifting. Bottom: the same network with the contact channel appended to its state vector places the block. The traces plot $|\delta|$ on the jaw joint; the contact plateau in both is the signal the top policy never receives. Both are computed from the same log fields.

16 of the 20 public datasets we targeted passed our frozen inclusion gates (Sec. 4.3, which also details the caveats on action semantics). The same signal supports runtime enforcement from bus telemetry alone (Sec. 4.4).

We ask what a small-data policy must observe to use this channel, and compare six observation designs that differ only in which function of the logged pair enters the state vector. The best tested form is the *lag excess* $e[t] = \delta[t] - \hat{K}r[t]$, which subtracts a per-joint free-motion following-lag model fit on the jaw-open frames of the training demonstrations, label-free and frozen before training (Sec. 3.1; implementation details in App. A).

Three results are established at our pre-declared seed resolution (Sec. 4.2). Channel-bearing observation beats channel-free in every input form. The tested auxiliary-prediction baseline recovers none of the direct-input gain: the same information moves performance as an input and not as a training target. And the lag excess is the only tested design to beat raw action history (26.6% vs. 13.8%, CI $[+2.8, +22.6]$); whether its edge over raw δ comes from the subtraction is not resolved at this budget, so we present it as the recommended instantiation, not a finding about the subtraction itself. On the public corpus, a pre-registered prediction fails: high δ at grasp time accompanies failed grasps, and the traces attribute this to operators over-commanding at contact (Sec. 4.3). A 40-line guard on the same telemetry removes 96% of simulated crush events without retraining (Sec. 4.4).

We summarize our contributions below:

- Characterization of servo tracking error as a contact/load proxy with a validity envelope and a resolution-vs-margin design rule (Sec. 3.2, 4.1).
- A pre-registered six-design observation study at 280 demonstrations: the channel helps as an input and not as an auxiliary prediction target (Sec. 4.2).
- A pre-registered 16-dataset corpus study associating high δ at grasp with proxy-defined failure (Sec. 4.3).

- A runtime jaw guard from bus telemetry alone, with paired deleted-success accounting (Sec. 4.4).

2 Related Work

Sensorless force estimation. Momentum-residual observers recover external torque on torque-controlled arms [9, 10]; virtual force sensors extend the idea to low-cost platforms [11], and Hwang et al. [12] estimate torque from commanded and measured positions on hobby servos, prior art for the raw channel; series-elastic actuation made deflection-as-force a design principle [13]. NeuralActuator [14] learns per-actuator models for external-force perception on this servo class, and FACTR 2 [15, 16] feeds estimated external torque into behavior cloning on commodity arms. Our lag excess is a deliberate special case of FACTR 2’s central idea, a linear free-motion baseline fit on positions alone; the restriction makes the signal retroactive, since every method above consumes live current or load telemetry, a calibration stage, or both. Closest in spirit, Wong et al. [8] show by intervention that ACT’s force competence depends on the leader-follower discrepancy implicit in its action labels; we place the signal explicitly in the observation and ask which form a small-data policy needs.

Observation design in imitation learning. ACT [1], Diffusion Policy [2], OpenVLA [4], π_0 [5], SmolVLA [6], and GR00T N1 [7] all condition on measured joint positions at one or two timesteps; none states a rationale for excluding actuator-error signals. The field’s argument against richer observations is causal confusion: past actions invite copycat shortcuts [17, 18, 19]. Our grid places both effects in one table.

Grasp monitoring and runtime safety. Industrial grippers ship grasp verification from actuator feedback as firmware: Robotiq’s object-detection register fires on commanded-vs-actual discrepancy under a current limit [20], and SCHUNK advertises workpiece-loss detection from an integrated encoder [21]; tactile alternatives require added sensing [22]. Academic safety filters for learned policies [23, 24, 25] consume a dynamics model, a state estimate, or exteroception; the guard of Sec. 4.4 applies the industrial telemetry-only approach to wrap learned policies, and we have not found a filter in the academic literature that runs from servo-bus telemetry alone.

3 Method

3.1 Problem formulation

Let $q[t]$ denote measured joint positions in integer encoder counts and $g[t]$ commanded goal positions, both logged at every control step (in LeRobot format, `observation.state` and `action`). The tracking error and lag excess are

$$\delta[t] = g[t-1] - q[t], \quad e[t] = \delta[t] - \hat{K} \cdot r[t], \quad r[t] = g[t-1] - g[t-2], \quad (1)$$

where $r[t]$ is the one-step command rate and $\hat{K}$ is a per-joint coefficient fit by least squares through the origin on the free-motion (jaw-open) frames of the training demonstrations, computed once and frozen before training (full computation, alignment convention, and normalization in App. A). Each servo runs a proportional position loop, so joint torque obeys $\tau \approx K_p \cdot \delta$ and jaw grip force obeys $F \approx \kappa \cdot \delta_{\text{jaw}}$ for a calibration constant κ in newtons per count. Two conditions make δ meaningful: the logged action must be the command actually applied to the servo, and command-to-state timing must be known and stable. Both hold by construction in our simulated plant; on real logs they are empirical properties of the recording stack (Sec. 4.3). We study which function of the logged pair (q, g) a policy trained on $N = 280$ demonstrations should observe.

3.2 The channel and its envelope

On our simulated plant the jaw calibration is $\kappa = 2.00$ N per count, linear while the jaw is seated and quasi-static, with held-out RMSE 4.3 N over 4–78 N. These constants describe the modeled plant:

the simulator implements the servo loop of Sec. 3.1, so its δ -to-force map is clean by construction; real STS3215 hardware adds stiction, backlash, deadband, saturation, and thermal drift that only bench calibration can measure (App. D). What simulation can characterize is the structure of the proxy: outside the seated quasi-static regime the signal is not a load reading, since free motion contributes following lag (which $e[t]$ removes) and fast transport contributes inertial deflection (App. B, Fig. 4). Encoder quantization yields a design rule: a safe-grip window of force margin M is learnable through the channel only if the observed quantum is finer than M/κ ; empirically the failure is a cliff whose location moves with the margin (Sec. 4.1).

3.3 System, task, and data

All experiments except Sec. 4.3 run on a simulated STS3215-class arm (five joints plus a one-DoF jaw, six servos) with a calibrated contact model. The task is pick-and-place of an object whose retention requires grip force above a lift threshold and whose integrity fails above a crush threshold. A privileged-force oracle passes an admission test on this task and, as a negative control, shows that standard rigid-block pick-and-place cannot distinguish force-aware from force-blind control at all. Demonstrations come from a scripted closed-loop teacher (280 episodes, success-filtered) whose grip regulator acts on the same bus-observable tracking error the arms study, with per-episode grip-depth targets drawn from $U[12, 45]$ counts and injected recovery perturbations; no privileged force sensing enters collection, the teacher’s commands are the applied commands by construction, and the observation arms see only functions of (q, g) . Policies are ACT [1], trained for 12,000 steps at 96 px, identical across arms; at this resolution we cannot rule out a camera deformation cue, but the base arm’s floor performance shows any such cue is insufficient alone. Seed variance was measured before any comparison: $\sigma_{\text{seed}} \approx 10$ points at $n = 100$ evaluation episodes, so 5-seed comparisons resolve gaps of roughly 11 points and we claim nothing finer. Evaluation episodes are common-random-number paired: every arm and seed rolls out the identical episode sequence.

3.4 Observation designs

Six arms share the training recipe and differ only in the proprioceptive observation. **Base (A)**. Measured positions $s[t]$ only: the stock LeRobot observation. **History (B)**. $s[t]$ plus the previous action $a[t-1]$: the minimal change that makes δ recoverable in principle, and the arm the copycat literature [17, 18] warns about. **Delta (C)**. $s[t]$ plus raw $\delta[t]$: the channel with its motion confound intact. **Auxiliary target (D)**. $s[t]$ only, with $\delta[t]$ predicted as an auxiliary target through a small head at loss weight 0.1: the same information as supervision rather than input. **Lag excess (E)**. $s[t]$ plus $e[t]$: the channel with free-motion lag subtracted. **Seat token (F)**. $s[t]$ plus a latched binary seat bit from the guard’s frozen stall detector run causally over $(s[t], a[t-1])$: the channel compressed to one bit.

4 Experiments

We evaluate: (i) over what envelope is δ a valid contact/load proxy; (ii) which observation design exploits it at small data; (iii) does the channel predict grasp outcomes on the public corpus?

4.1 Channel behavior on the simulated plant

The calibration of Sec. 3.2 holds in the seated quasi-static regime. Because the simulator implements the servo loop that generates δ , these numbers validate the proxy against our own plant model; their value is in mapping where the proxy’s structure breaks. A stall detector on net travel and lag excess localizes seat events, and constructed adversarial negatives (free-space reversals, transport shakes) do not fire it within the declared envelope. Coarsening the observed channel collapses force-aware grasping once the quantum exceeds M/κ , and rerunning the sweep at a different margin moves the cliff accordingly.

Table 1: Placement success (%), mean $\pm$ std over 5 seeds (500 paired episodes per arm), with differences against the channel-free base (A) and the action-history baseline (B). Brackets are hierarchical-bootstrap 95% CIs (resampling seeds, then episodes within seeds); an interval containing zero is unresolved at this budget. Best in bold.

Arm	Observation	Success (%)	vs. A	vs. B
A base	$s[t]$	6.2 ± 5.9		
B history	$s + a[t-1]$	13.8 ± 7.7	$+7.6 [-0.8, +16.0]$	
C delta	$s + \delta$	21.8 ± 9.7	$+15.6 [+5.6, +25.2]$	$+8.0 [-2.8, +18.6]$
D aux-target	$s; \delta$ predicted	6.6 ± 4.0	$+0.4 [-6.2, +6.4]$	
E lag excess	$s + e$	26.6 ± 7.8	$+20.4 [+11.6, +29.0]$	$+12.8 [+2.8, +22.6]$
F seat token	$s + \text{seat bit}$	21.8 ± 9.5	$+15.6 [+6.0, +25.2]$	$+8.0 [-2.6, +18.6]$

4.2 Observation design at 280 demonstrations

Table 1 and Fig. 2 give the grid: 6 arms $\times$ 5 seeds $\times$ 100 paired evaluation episodes, identical otherwise. Arm definitions and directional predictions were frozen before implementation, with E vs. B declared the headline comparison; frozen documents, per-seed outcomes, and all runs are in the released repository. Alongside Welch’s t across seed means we report 95% CIs from a hierarchical bootstrap resampling seeds, then episodes.

Three results resolve. First, channel-bearing observation beats channel-free in every input form: C, E, and F each clear the base by at least 15 points with intervals excluding zero (Table 1). Second, the tested auxiliary-prediction design does not recover the direct-input gain: arm D lands equal to base while the same information as an input triples success; other auxiliary weightings and architectures could in principle close this gap, and none was tuned here. Third, in the declared headline comparison, only the lag-excess form beats action history at resolution, by 12.8 points (Welch’s $t = 2.6$ across seed means); raw δ ’s 8-point margin over history does not resolve. The remaining pairs stay open, including E against C (difference $+4.8$, CI $[-6.2, +15.8]$), so whether E’s edge comes from the subtraction or from δ at all is not attributable at this budget; resolving $E-C$ by seed expansion would require roughly 40–60 seeds per arm at this σ_{seed} , and evaluation is already episode-paired. The frozen prediction named E the sharpest arm and expected D, E, and F to clear B; D’s share of that prediction failed, as reported above. An interim 3-seed analysis flagged E’s margin as resolved and F’s as at-the-floor, and the expansion to 5 seeds confirmed both calls, collapsing F’s margin over B and preserving E’s. Caveats: five seed means per arm, a headline difference of similar magnitude to the declared resolution, and uncorrected secondary comparisons beyond the designated primary one. Within those limits the ordering supports one design statement: at small data on this force-critical task, observing the channel is the change that moves performance, and the free-motion-compensated form is the best tested way to observe it.

4.3 Ecosystem study: a pre-registered negative

Does δ predict grasp outcomes corpus-wide? The protocol was frozen before the run: seat detection by the stall detector of Sec. 4.1, δ -at-seat as unsigned tracking error, success proxy defined as the gripper staying closed through transport, and the prediction that successes carry higher δ with Cohen’s $d > 0.5$ in at least 10 of 20 datasets. Of the 20 targeted SO-100/SO-101 datasets, 16 passed the frozen inclusion gates with connectivity: 546 episodes, 283 proxy successes.

The prediction fails: 2/16 datasets (pooled within-dataset $d = -0.39$). The channel is informative with the opposite sign: $|d| > 0.5$ in 9/16 datasets, 6/16 significantly inverted (Fig. 3). The traces explain the physics. Teleop leader devices have no hard stop, so operators over-command every close; the seat flag fires in nearly every episode of 14/16 datasets because empty jaws stall at the mechanical limit exactly as loaded ones do. Successful grasps are stereotyped low- δ closes (in one dataset every clean episode rests at the same 7-count gap), while failures carry 40–80-count frames of an operator forcing a jammed grasp. High δ at grasp time is associated with failure under the

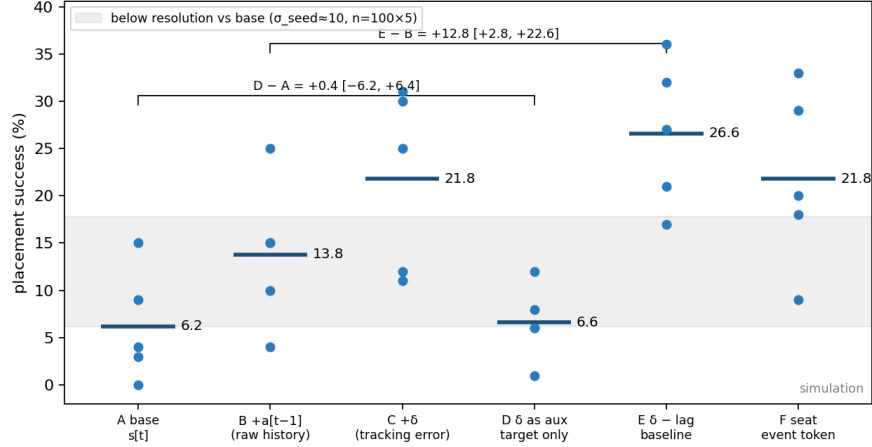


Figure 2: The observation-design grid. Bars are 5-seed means, dots individual seeds (100 episodes each). The gray band spans the below-resolution zone around base given $\sigma_{\text{seed}} \approx 10$. The lag-excess arm (E) is the only design whose margin over the history baseline (B) exceeds seed resolution.

188 frozen proxy. This rules out mining the corpus for success labels through this channel alone, and it
 189 supports enforcement: the high- δ events the corpus associates with failure are what the guard caps.

190 The result also bounds Sec. 4.2’s transfer to retroactive training on real teleop data. When the seat
 191 event fires in essentially every episode, a latched seat bit (arm F) carries no information, and raw
 192 δ at close time reads commanded depth as much as contact; what stays informative is δ magnitude
 193 and its dynamics, which is what the inverted effect sizes measure. Whether the lag-excess form
 194 retains its simulated advantage when trained on over-commanded human demonstrations is an open
 195 empirical question, not a corollary. Two caveats attach to the corpus numbers: the recorded action
 196 is the leader position, which approximates but need not equal the applied follower command, and
 197 action-to-state timing is characterized only up to an integer-offset sweep over $k \in \{0, \dots, 5\}$, which
 198 leaves these conclusions unchanged (the median per-dataset shift in Cohen’s d across offsets is 0.12)
 199 while estimating the physical alignment at three to four recorded frames (App. A).

200 4.4 Runtime guard

201 The same telemetry supports enforcement. A 40-line jaw guard (net-travel stall and lag-excess seat
 202 detection, a 12-count squeeze budget past seat, a closing-rate limit) runs from bus signals with no
 203 dynamics model, no retraining, and no object knowledge; all results are simulated. Paired its detec-
 204 tion thresholds make it insensitive to closes slower than its rate gates, so it engages on aggressive
 205 closers and not on the gentle scripted teacher (App. C), and on three learned checkpoints over iden-
 206 tical paired episodes its first-generation detector (App. C) reduced crush events from 85 to 3 of 180
 207 (96%), with median peak force dropping from 33–131 N to 13–26 N; it also deletes successes one
 208 policy earned by over-gripping. Demonstrations must record applied commands, and wrapped con-
 209 trollers must be closed-loop on contact. Details, figure, and integration accounting in App. C; bench
 210 calibration and hardware validation are in progress (App. D).

211 5 Limitations

212 All learning and guard results are simulation, on one gripper and one object family; every quantity in
 213 newtons inherits the simulator’s servo and contact model until bench calibration lands (App. D). The
 214 task is constructed to require force-band control, so the grid measures observation design on a force-
 215 critical task, not manipulation in general. The lag-excess model is a first-order correction; richer
 216 free-motion predictors [15] may compensate better and are untested. The seed budget leaves $C-B$

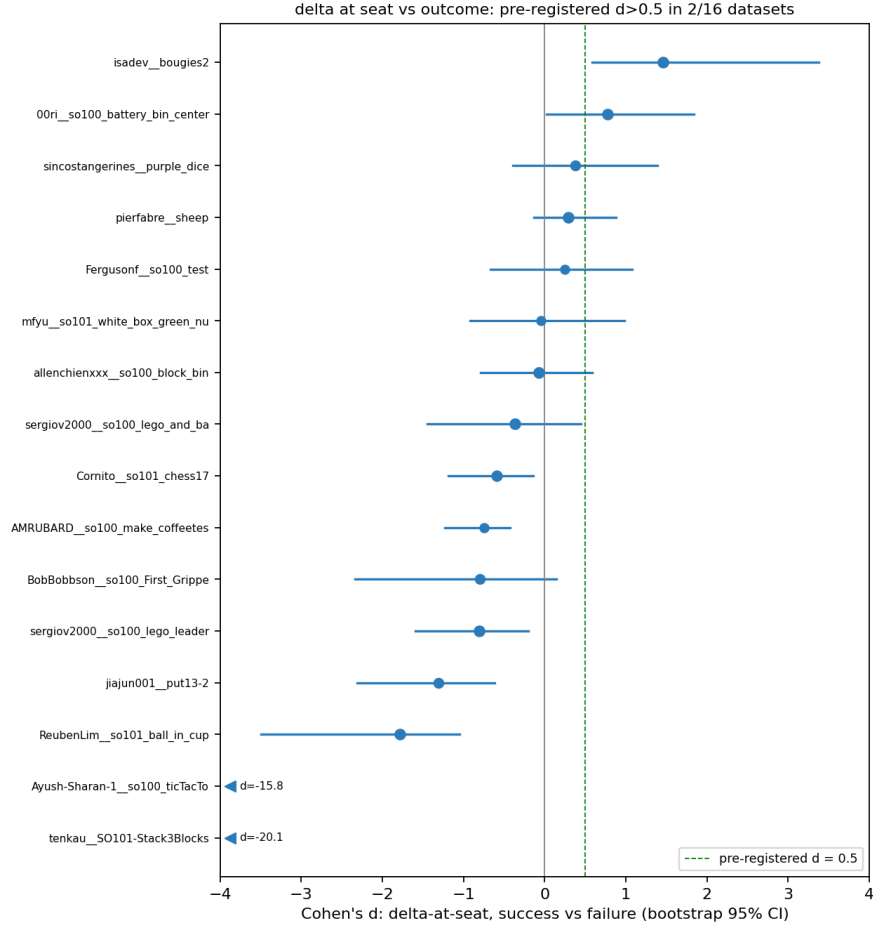


Figure 3: Per-dataset Cohen’s d of δ -at-seat between proxy successes and failures, bootstrap 95% CIs, for the 16 datasets passing the frozen gates (attrition documented in the repository). Positive d was the pre-registered direction. Two datasets exceed $d = 0.5$ as predicted; six are significantly inverted; two task-repetitive datasets reach $d \approx -16$ and -20 because their stereotyped successes have near-zero δ variance.

and $E-C$ open, so the value of the subtraction itself is not established. On real logs, δ depends on action semantics and timing that the corpus study characterizes only up to an integer-offset sweep. The corpus success proxy is unvalidated against ground truth, and multi-grasp tasks were excluded by design. The grid is measured on one demonstration collection; a three-seed recollection (App. C) surfaces collection-level resampling variance that our comparisons do not yet average over. The channel is valid only seated and quasi-static.

6 Conclusion

Position-command logs contain a retrospectively computable contact proxy: the servo tracking error, valid within a quasi-static envelope our simulated plant makes measurable. At 280 demonstrations, policies must observe this channel; predicting it as the tested auxiliary target recovers none of the gain, and the free-motion-compensated form is the best tested instantiation. Across 16 public datasets high δ at grasp time accompanies failure, which rules out free outcome labels from this channel alone and supports its use for enforcement. What remains open requires hardware: bench calibration, retroactive training on real teleop logs, and a copycat-style probe of when history helps versus hurts in behavior cloning.

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296 A Channel computation and implementation details

297 All signals are in integer encoder counts at the recorded cadence (stride-2 frames of a 30 Hz bus,
 298 i.e. 15 Hz): $\delta_j[t] = a_j[t-1] - s_j[t]$, where a is the commanded and s the measured position, the
 299 $k=1$ alignment convention. An offset sweep over $k \in \{0, \dots, 5\}$ on the public corpus leaves
 300 every Sec. 4.3 conclusion unchanged (the median per-dataset shift in Cohen’s d across offsets is
 301 0.12) while estimating the physical alignment at three to four recorded frames; the bench latency
 302 measurement (App. D) will pin it. The commanded rate is the first difference of commands, $r_j[t] =$
 303 $a_j[t-1] - a_j[t-2]$, zero at the first two frames of an episode. The lag model is a single per-joint
 304 scalar through the origin, $\hat{K}_j = \sum r\delta / \sum r^2$, fit over frames whose jaw command exceeds its 30th
 305 percentile; the jaw is the task’s only contact, so jaw-open frames are free motion. Fitted values
 306 span $\hat{K} = 0.74\text{--}0.84$ across joints on our training set (and move by at most 0.09 when the free-
 307 motion percentile criterion is varied from the 10th to the 50th), computed once on the training
 308 demonstrations and frozen. The policy input is the concatenation $[s_{\text{norm}}, (\delta - \hat{K}r - \mu_\delta)/\sigma_\delta]$; joint
 309 states and actions are standardized by dataset statistics, while δ , e , and the aux target use a fixed scale
 310 of $\sigma_\delta = 8$ counts with $\mu_\delta = 0$, so no channel statistic leaks from evaluation. The same statistics
 311 are saved beside the checkpoint and applied identically at evaluation, where $a[t-1]$ is the previously
 312 issued command, so the feature is computable from exactly what the deployed stack already has.
 313 Appended channels concatenate onto the state vector, leaving the network otherwise unchanged.

314 B Mechanism on a single grasp

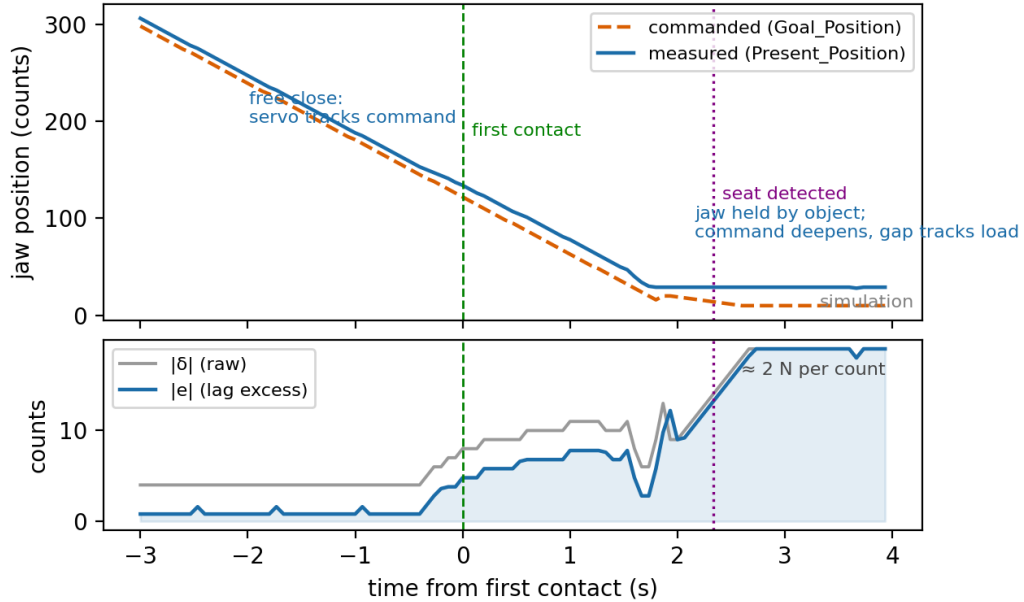


Figure 4: The channel on one servo and one grasp. The commanded position (dashed) and measured position (solid) track closely through the free close, separated only by the small following lag. At first contact (marked) the measured position stalls while the command continues to deepen; the growing gap is the load reading, $\kappa \approx 2$ N per count on the simulated plant. The detector’s seat event (marked) fires when net travel stalls while lag excess grows. The lower panel plots $|\delta|$ and the lag excess $|e|$ over the same frames; $|e|$ sits near zero through the free close by construction and inherits the load reading after contact.

315 C Guard details

316 The guard combines net-travel stall and lag-excess seat detection, a squeeze budget of 12 counts
 317 ($\approx +24$ N on the simulated plant) past seat, and a closing-rate limit. Two regimes follow from
 318 the detector’s thresholds. On aggressive closers it binds: learned policies commanding fast or deep
 319 closes are capped near 40 N against raw peaks of 99–127 N, and an open-loop clamp reference com-
 320 manding 360 N grips is held to 10–28 N. On the gentle closed-loop teacher (~ 0.3 counts per frame in
 321 its fine pass, below both detection gates) it never fires: a paired run shows 25/30 placements wrapped
 322 versus 28/30 raw at statistically indistinguishable ~ 39 N grips, i.e. no enforcement occurred and the
 323 run measures wiring overhead only. Whether threshold-level insensitivity to gentle closes is accept-
 324 able conservatism or a coverage gap depends on the deployment’s hazard model; we state it as a
 325 measured property of these thresholds.

326 An independently recollected demonstration set (same teacher, recipe, and budget; fresh episodes)
 327 trained at three seeds per arm neither resolves any per-arm difference from the original grid nor re-
 328 produces its point-estimate ordering at that reduced seed count (the lag-excess arm’s fresh-collection
 329 seeds span 7–27 against the original 17–36, Welch’s $t = 1.6$). Across the six arms the three channel-
 330 bearing designs’ point estimates fell while the three channel-free designs’ rose, none individually re-
 331 solving; four marginal statistics of the two collections (tracking-error distributions, command floors,
 332 command-rate distributions, per-episode grip-depth diversity) are indistinguishable, so we treat this
 333 as unattributed collection-level resampling variance – a component above seed variance that the
 334 headline comparison of Sec. 4.2 does not yet average over, and that hardening it requires spanning
 335 with seed expansion over multiple collections. The crush-elimination A/B predates the detector’s
 336 final revision: on three learned checkpoints over identical paired episodes the first-generation de-
 337 tector (single-step stall test, 4-count budget) reduced crush events from 85 to 3 of 180 across crush
 338 tiers (Fig. 5), with median peak force dropping from 33–131 N to 13–26 N; a rerun under the final
 339 detector is pending alongside hardware validation. The guard also deletes successes a policy earned
 340 by over-gripping: one quantized arm lifted 7/30 episodes at a 131 N median, and those successes
 341 vanish under the cap. Integration requires two properties: demonstrations must record applied com-
 342 mands rather than intentions, and any wrapped controller must be closed-loop on contact, since the
 343 rate limiter breaks open-loop grasp timing by construction. Collecting demonstrations under *active*
 344 guard enforcement would require a detector variant sensitive to gentle closes and remains untested.

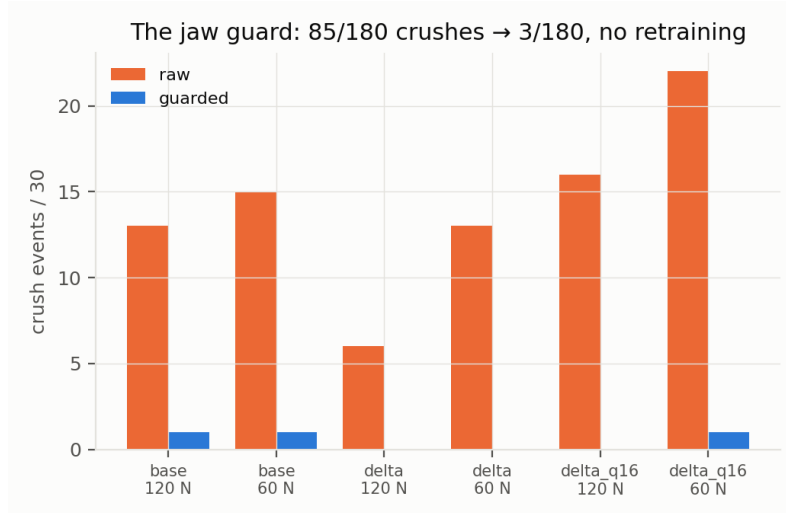


Figure 5: Crush events per 30 paired episodes for three learned checkpoints (base, delta, and a 16-count-quantized delta arm) at two crush tiers, raw (orange) versus guarded (blue), under the first-generation detector. Identical checkpoints and episode seeds; the guard acts only through bus telemetry at runtime. Totals: 85 of 180 raw, 3 of 180 guarded.

345 **D Hardware calibration protocol (in progress)**

346 δ versus Present_Load, Present_Current, and a load cell on a physical STS3215; staircase pro-
347 tocol at three speeds, with opening and closing directions, repeated trials, session-start/middle/end
348 drift checks, and a command-to-state latency distribution measurement. Comparison rows: the sim-
349 ulator’s 2.0 N/count, Hwang et al. [12], and NeuralActuator [14]. Until this lands, every newton in
350 this paper inherits the simulator’s contact model; a negative transfer result would itself be the first
351 measurement of the channel’s sim-to-real gap.